

Effect of Solder Paste Alloy and Volume on Solder Voiding

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Abstract— Lead-free solder alloys have been extensively utilized and examined since the early 2000s, owing to the harmful effects of lead on the environment and health. However, it is argued that lead-free solder joints are more susceptible to solder voids, lowering their reliability. Solder voids are cavities or holes that indicate a lack of solder alloy in some areas of the solder joint and may include air, flux, and cured resins. This study explored the impact of solder paste alloys, solder paste volume, and surface finishes on the number and size of solder voids after reflow. The test vehicle of this study consists of a chip array ball grid array (CABGA208) and 2512 surface mount resistors (SMR) components assembled with three solder paste alloys (SAC305, SAC-3.0 Bi, and SAC-3.2 Bi-3.0 Sb). On the boards, three different paste volume ratios (0.1, 0.3, and 0.5) were printed with two surface finishes, including electroless nickel immersion gold (ENIG) and organic solderability preservative (OSP). Mountains9 software was used to capture and analyze X-ray images of CABGA208 components to evaluate solder voiding. The results demonstrated that the size of the voids increased proportionately to the paste/sphere volume ratio. Among solder paste alloys, SAC-3.2 Bi-3.0 Sb exhibited the highest voiding, whereas SAC305 and SAC-3.0 Bi had the best size characteristics. Furthermore, in most combinations, OSP revealed larger voids than ENIG.

Keywords— Paste Volume, Voids, Lead-free, surface finish, solder joints

I. INTRODUCTION

Concerns regarding the environmental and health implications of lead exposure have prompted the electronic industry to study lead-free alloys to replace lead alloys throughout the last decade. Sn-Ag-Cu alloys, which comprise tin, silver, and copper, have been proposed for use in electronics and other applications requiring high reliability owing to their excellent electrical and thermal conductivity, mechanical properties, and corrosion resistance. To improve the reliability of these alloys, dopants such as indium (In), bismuth (Bi), and antimony (Sb) were added [1]–[3]. However, these new alloys have various reliability concerns, one of which is solder voiding [4]–[6]. Voids in solder joints, which are places where the solder has not sufficiently filled the joint and left a gaping space. This could occur if there is insufficient solder or if the solder does not adequately wet the bonded surfaces. The presence of voids in a solder joint increases the risk of

mechanical and electrical failures. Other reliability challenges induced by voids include solder bridging and poor thermal conductivity [7]–[10]. In order for the voids area to be considered acceptable by the IPC-7095 standard, it must be less than 25% of the actual x-ray image area [11].

Solder voids can be divided into many categories, including macro-voids, planar micro-voids, and shrinkage voids [7]. This study focuses on macro voids, which are the most common type of voids. Macro voids, also known as process voids, are produced by volatile outgassing elements in fluxes and solder pastes [7], [12]. Furthermore, flux is a chemical cleaning agent added to solder paste to remove contaminations and oxides from the soldering region. It is a volatile element that begins to evaporate during the soldering process. However, some of these compounds get trapped and produce voids within the solder joint [13], [14]. The process of void formation is illustrated in Fig. 1.

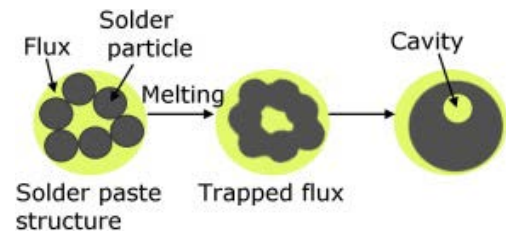


Fig. 1. Void Formation Process [15]

Flux is triggered at a specific temperature during the reflow process to remove oxides from solder particles and soldered surfaces [15]–[17]. Because the quantity of oxides is affected by the paste alloy and paste volume, it was inferred that the solder pastes composition and volume substantially affected void formation [18], [19]. Furthermore, increasing the paste volume indicates that more gaseous elements will be detected, which means more compounds will be trapped and voids will develop [7].

The surface finish type also influences the formation of the void [7]. Bernard and Bryant [20] evaluated the influence of several surface finishes on solder voiding and observed that OSP voids are larger than those of other surface finishes tested.

In recent years, many studies have been conducted to investigate the void impact on lead-free solder joints. Rauer et al. [21] utilized x-ray computed tomography (CT) to assess void content, locate voids, and estimate their volumes in lead-free solder joints. The results showed that macro voids could be the source of crack initiation. The results showed that macro voids could be the source of crack initiation. Steiner et al. [22] investigated the influence of soldering profile, void formation, and mechanical strength on solder joint reliability while employing various paste alloys and surface finishes. They concluded that surface finishes, soldering profiles, and solder alloys have an impact on void formation.

Overall, the void influence on lead-free solder joints has serious consequences for electronic device reliability and performance. Therefore, further studies are required to fully comprehend the influence of solder paste alloy, solder paste volume, surface finish, and solder joint location on solder voiding. This study examines the impact of solder paste alloys, solder paste volumes, surface finishes, and solder joint locations on the size of the voids. X-ray images are acquired for each CABGA208 component. These images were analyzed using Mountains9 software, which detects voids within the joints. This study offers a novelty in its focus on these specific factors that affect solder voiding, contributing to a better understanding of the relationship between these factors and solder void formation.

II. EXPERIMENTAL PROCEDURE

A. Assembly Process

The test vehicle used in this study includes three chip array ball grid array (CABGA208), and six Surface Mount Resistors (SMR2512) mounted on an FR-R glass-epoxy substrate with four layers. Two distinct surface finishes were used, which are ENIG and OSP. To mount CABGA208 components, three solder paste alloys, namely SAC305, M758, and M794, were screen printed using three stencils to obtain varied solder paste volumes along with 18 mils SAC305 solder spheres, as indicated in Table 1. The composition of these solder alloys is listed in Table 2.

TABLE I. STENCIL THICKNESS AND PASTE VOLUME

Paste Volume	Stencil Thickness (mil)	Paste to sphere volume ratio for CABGA208	Aperture diameter (mil)
Low	4	0.1	10
Medium	5	0.3	15.3
High	6	0.5	18

TABLE II. SOLDER ALLOY COMPOSITION

Solder Alloy	Composition
SAC305	Sn - 3.0Ag - 0.5Cu
M758	Sn - 3.0Ag - 3.0Bi - 0.8Cu
M794	Sn - 3.4Ag - 3.2Bi - 3.0Sb - 0.7Cu

This assembly is reflowed with the reflow profile depicted in Fig 2. The reflow profile consists of preheating to 150 °C in 200 sec and a soak from 120 °C to 150 °C in 120 sec. The peak temperature of this profile is 250 °C at 60 sec above the liquidus. After that, it cools at a rate of 3.5 °C per second.

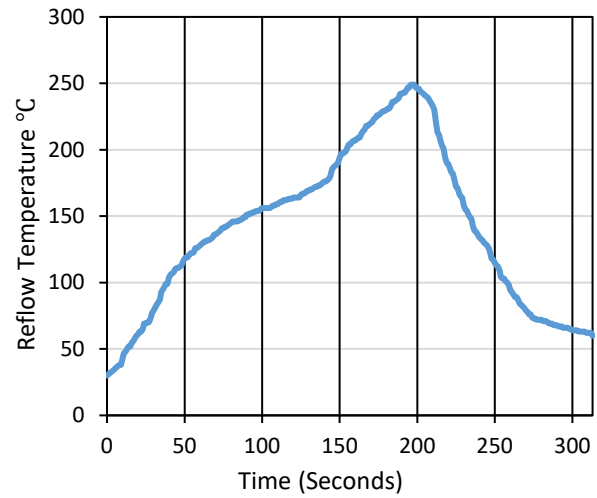


Fig. 2. Reflow Profile

The microstructure study was performed on the component shown in Fig. 3, while the void analysis was performed on the other two components. Consequently, two components of each assembly combination were analyzed in this study for a total of 36 parts, as presented in Table 3.

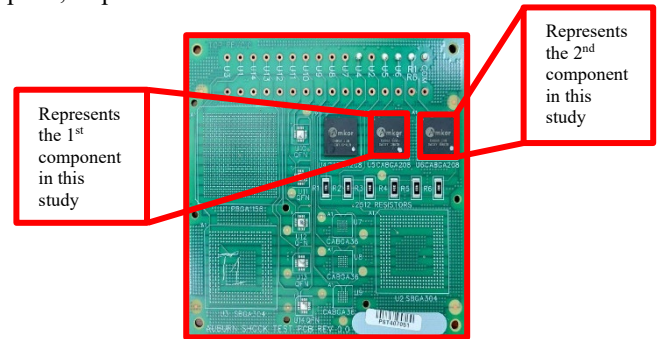


Fig. 3. Test Vehicle

TABLE III. TEST MATRIX

Solder Paste Alloy	Volume ratio	Surface Finish
SAC305	Low	ENIG OSP
	Medium	
	High	
M758	Low	
	Medium	
	High	
M794	Low	
	Medium	
	High	

B. Void Analysis

For the two CABGA208 components under consideration for this study, X-ray images were captured. Fig. 4 illustrates a sample of one combination.

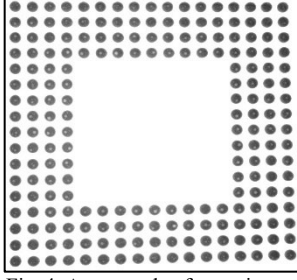


Fig. 4. An example of x-ray image

Consequently, the contrast of these X-ray images has been adjusted using ImageJ software. After processing, voids were identified and analyzed using Mountains9 software. This software also enables users to inspect each joint separately and identify voids. Fig. 5 shows an example of a joint that was evaluated using Mountains9. Moreover, Mountains9 detects all voids inside a solder joint and provides their area, and then the total voids area of each solder joint has been computed.

In this study, one side and one corner of each component's solder joints were inspected and analyzed, as depicted in Fig. 6.

The voids area within each joint was exported from the software, and the total voids area of each joint was determined. Following that, the average voids area per joint was computed by dividing the total voids area of each component by the number of joints tested, 52 joints. Based on these results, the influence of paste alloy, paste volume, surface finish, and detection region (side and corner) on the voids area has been explored.

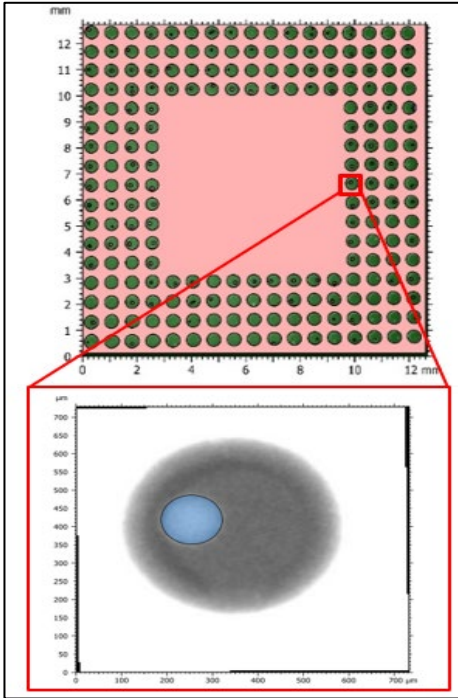


Fig. 5. An example of a void detected inside a joint by Mountains9

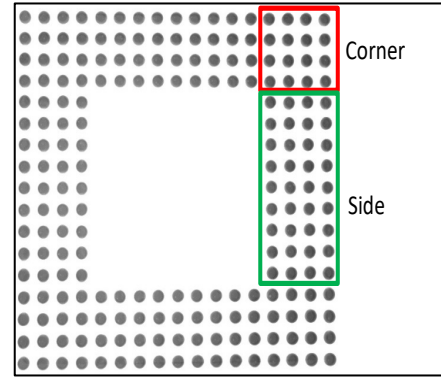


Fig. 6. Solder joint locations investigated

III. RESULTS AND DISCUSSION

ImageJ and Mountains9 software were used to modify and analyze the x-ray images of 36 CABGA208. Furthermore, 52 joints from each component were analyzed to identify voids and measure their area. The total voids area of each component was divided by 52 to produce the average voids area per joint, which was employed in this study.

A. Effect of Solder Paste Volume

Table 4 displays the average voids area per joint ($\mu\text{m}^2/\text{joint}$) for CABGA components printed with different volumes of SAC305 paste and OSP surface finish. The results indicate that the voids area was the most for high paste volume, followed by medium and low paste volume. This pattern can be observed in both of the components examined. This is attributed to a higher paste volume associated with increased flux and gaseous elements that might get trapped within the joints and generate voids [15].

TABLE IV. VOIDS AREA OF SAC305 WITH OSP.

Paste Volume	Avg. Voids Area / Joint ($\mu\text{m}^2/\text{joint}$)	
	OSP	
	1 st Component	2 nd Component
High	7,496	7,768
Medium	4,831	3,870
Low	1,557	1,043

When Bi is added to the SAC alloy, the situation changes slightly. Table 5 demonstrates that the voids are more prominent in the medium paste volume than in the high paste volume, although the difference is minor. However, the voids of high and medium paste volumes are much higher than those of low paste volumes.

TABLE V. VOIDS AREA OF M758 WITH OSP

Paste Volume	Avg. Voids Area / Joint ($\mu\text{m}^2/\text{joint}$)	
	OSP	
	1 st Component	2 nd Component
High	6,442	7,909
Medium	6,791	9,226
Low	3,646	4,294

Regarding M794, which contains 3.2% Bi and 3.0% Sb, the voids area increases proportionately with paste volume, as illustrated in Table 6.

TABLE VI. VOIDS AREA OF M794 WITH OSP

Paste Volume	Avg. Voids Area / Joint ($\mu\text{m}^2/\text{joint}$)	
	OSP	
	1st Component	2nd Component
High	15,076	12,766
Medium	11,033	9,783
Low	2,101	1,860

Therefore, it can be deduced that the volume of the paste has a substantial effect on the void area. As demonstrated in Figure 7, the ENIG surface finish exhibits the same trend and influence of the solder paste volume regardless of the alloy used.

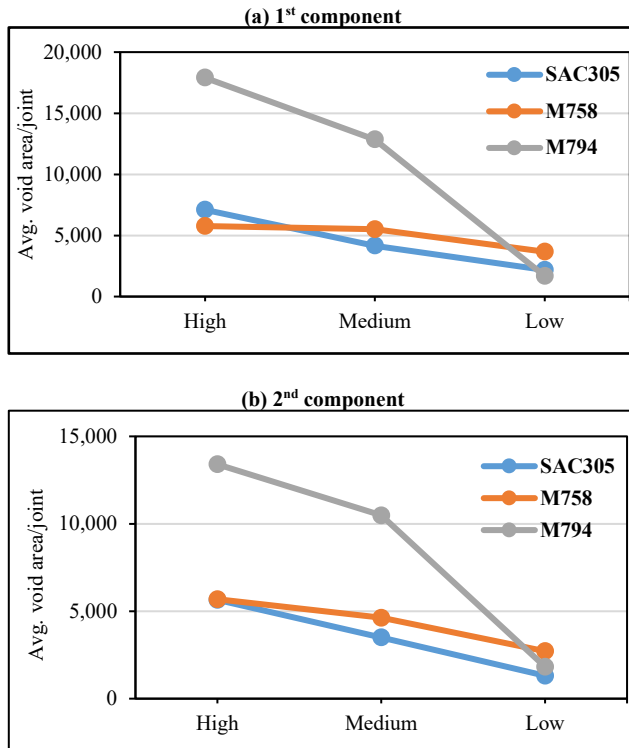


Fig. 7. Voids area of ENIG for the (a) 1st component and (b) 2nd component

B. Effect of Surface Finish

Fig. 8 depicts the influence of surface finish type on solder voiding under varying SAC305 volumes. It is evident that OSP has larger voids than ENIG in all combinations, except for the low volume when ENIG has the highest void volume. OSP consists of an organic substance that evaporates during soldering and produces gases that exacerbate voids [7].

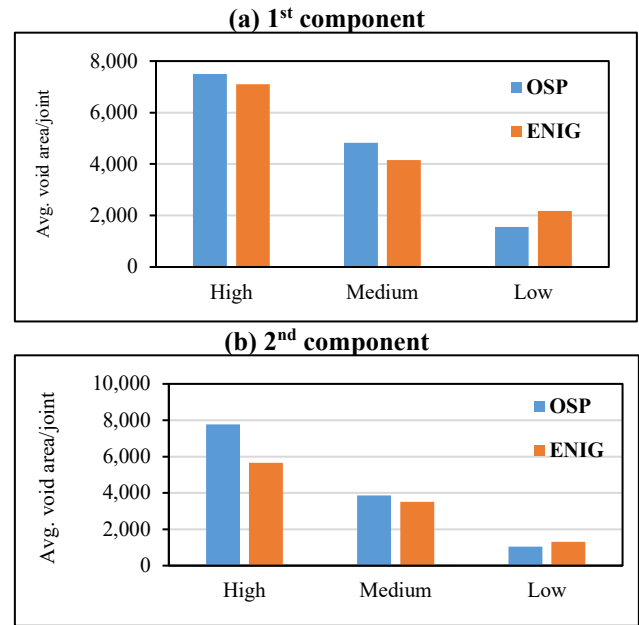


Fig. 8. Voids area of SAC305 for the (a) 1st component and (b) 2nd component

In terms of the difference between OSP and ENIG in both components, Fig. 9 demonstrates that M758 performs identically to SAC305. In all combinations, the void volume of OSP exceeds that of ENIG.

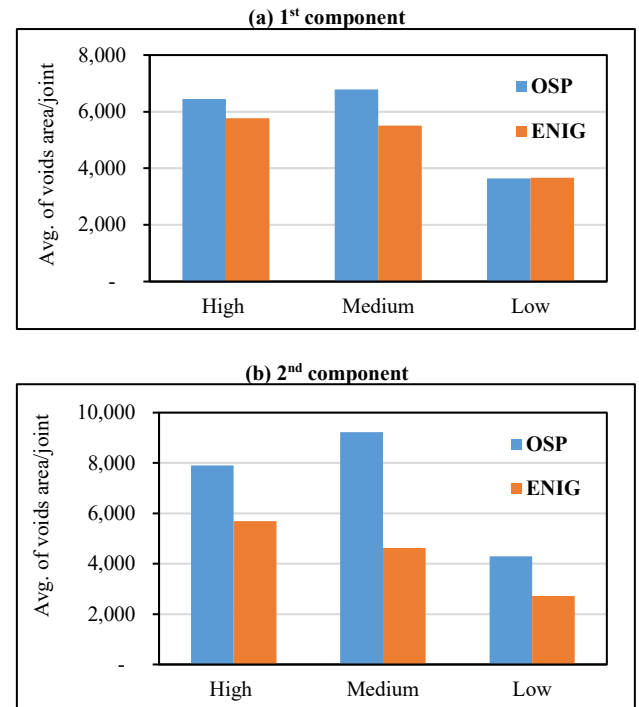


Fig. 9. Voids area of M758 for (a) 1st component and (b) 2nd component

ENIG surface finish with M794 had a more substantial influence than OSP on solder voiding in most of the combinations of both components, contrary to the findings of

SAC305 and M758. In addition, Fig. 10 demonstrates that OSP performs better than ENIG in terms of void area for both high and medium paste volumes. However, the reduced paste volume indicates that ENIG has a smaller void area.

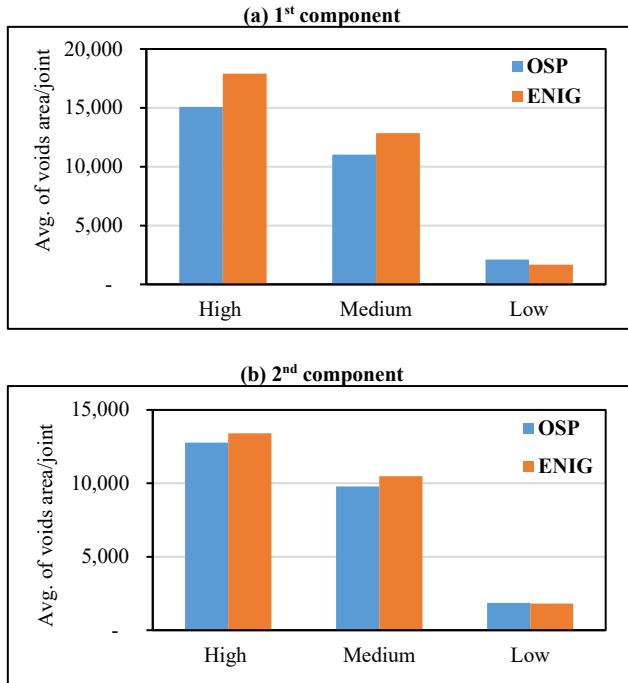


Fig. 10. Voids area of M794 for the (a) 1st component and (b) 2nd component

As a result, ENIG outperformed OSP, although not in all combinations. This is because OSP contains an organic ingredient that evaporates during the soldering process, resulting in more gases and the formation of more voids. [7]. As a result, statistical analysis is required to demonstrate that.

C. Effect of Solder Paste Alloy

The high and medium volumes of M794 had considerably larger voids than SAC305 and M758 in the two components because of the presence of dopants, specifically 3.2% Bi and 3.0% Sb, as shown in Fig. 11. Furthermore, Sb has a high coefficient of thermal expansion at room temperature, with a value of approximately 51 ppm/°C, which implies that it expands more when it is heated during the reflow. This high expansion stimulates the formation of voids. However, M758 has a slightly larger voids area with a lower paste volume. In terms of voids area, SAC305 and M758 perform almost identically.

D. Effect of Solder Joint Location

The influence of solder joint location was also evaluated, as the corner and side illustrated in Fig. 6 were compared in terms of average voids area per joint. Table 7 indicates that solder joints printed on the side using SAC305 solder paste with OSP surface finish have larger voids than those printed on the corner, regardless of paste volume in most combinations.

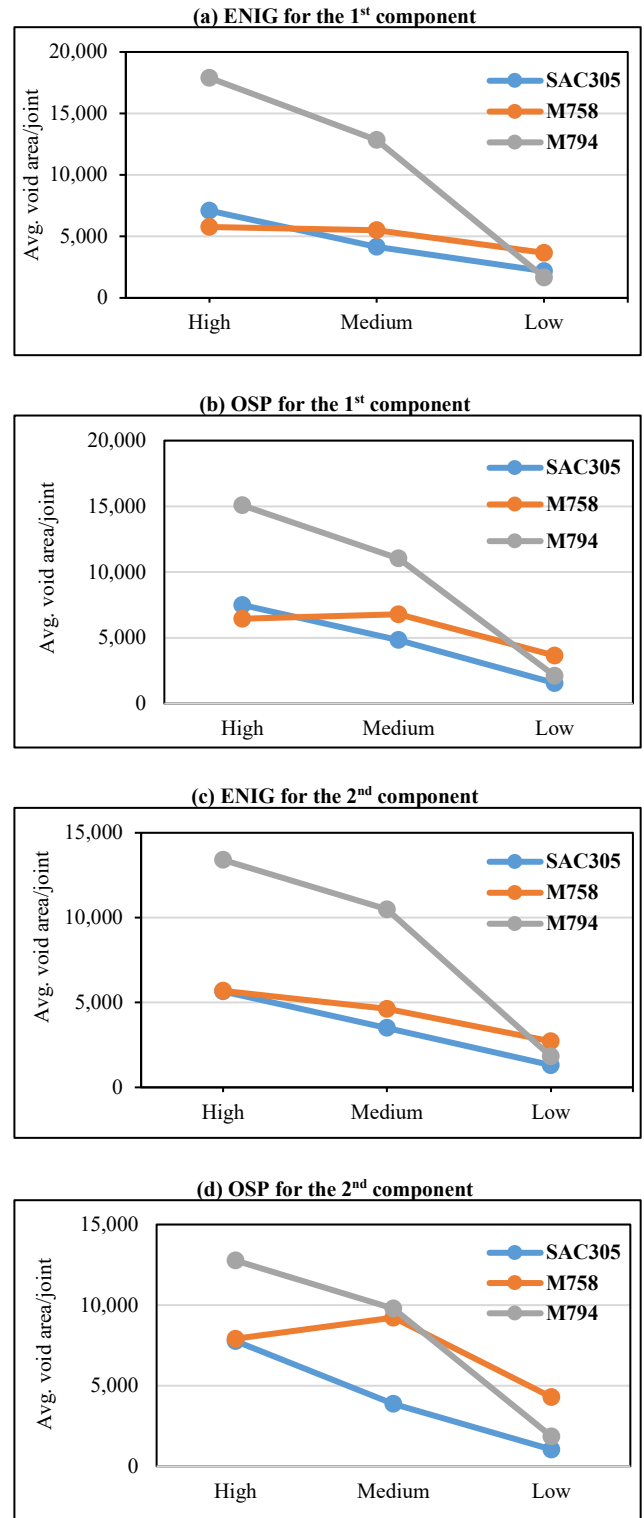


Fig. 11. Voids area of (a) ENIG and (b) OSP for the 1st component; Voids area of (c) ENIG and (d) OSP for the 2nd component.

TABLE VII. VOIDS AREA OF SAC305 AND OSP ON CORNER AND SIDE.

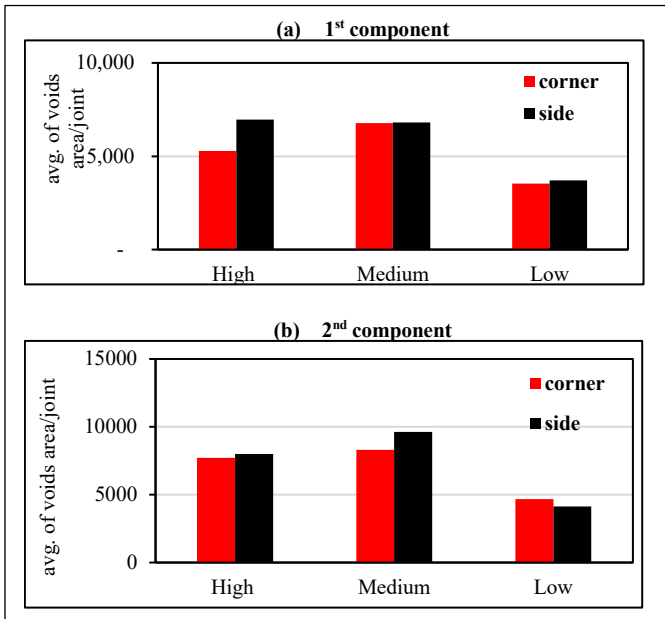
Paste Volume	Avg. Voids Area / Joint ($\mu\text{m}^2/\text{joint}$)			
	Corner		Side	
	Component 1	Component 2	Component 1	Component 2
High	7,254	8,742	7,603	7,335
Medium	4,634	4,328	4,919	3,667
Low	1,761	1,119	1,466	1,009

On the other hand, the side has larger voids than the corner in solder joints of high SAC305 with ENIG, while the corner occupied the top with the medium and low paste volume, as shown in Table 8.

TABLE VIII. VOIDS AREA OF SAC305 AND ENIG ON CORNER AND SIDE.

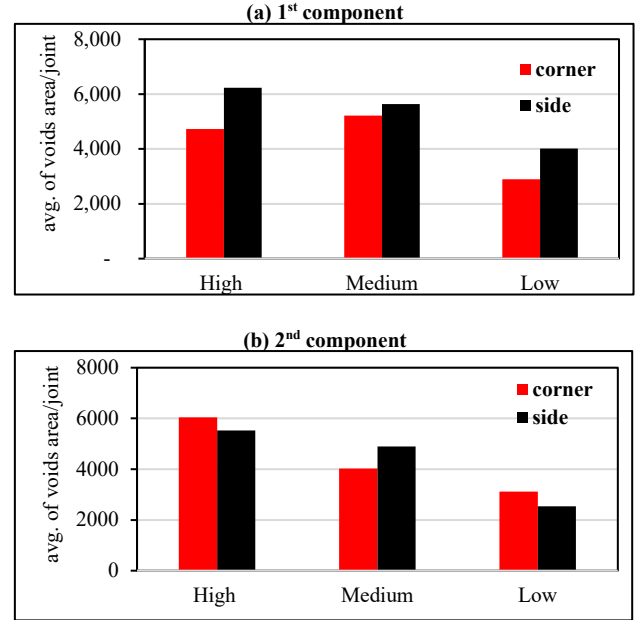
Paste Volume	Avg. Voids Area / Joint ($\mu\text{m}^2/\text{joint}$)			
	Corner		Side	
	Component 1	Component 2	Component 1	Component 2
High	6,579	4,033	7,326	6,369
Medium	4,358	4,198	4,061	3,194
Low	2,904	1,491	1,850	1,230

As shown in Fig. 12, the results of M758 solder paste with OSP show that the side has more voids than the corner in most combinations of both components.

Fig. 12. Voids area of M758 and OSP on corner and side for the (a) 1st component and (b) 2nd component

The results of M758 with ENIG surface finish are summarized in Fig. 13. In the first component, it was observed that the side contains more voids than the corner. Nonetheless,

the second component exhibits the inverse in high and low paste volumes.

Fig. 13. Voids area of M758 and ENIG on corner and side for the (a) 1st component and (b) 2nd component

Finally, regardless of surface finish type or paste volume, the side has larger voids than the corner with M794 solder paste, as demonstrated in Tables 9 and 10.

TABLE IX. VOIDS AREA OF SAC305 AND OSP ON CORNER AND SIDE

Paste Volume	Avg. Voids Area / Joint ($\mu\text{m}^2/\text{joint}$)			
	Corner		Side	
	Component 1	Component 2	Component 1	Component 2
High	13,442	11,510	15,802	13,324
Medium	10,993	9,299	11,050	9,999
Low	2,283	1,777	2,020	1,896

TABLE X. VOIDS AREA OF SAC305 AND ENIG ON CORNER AND SIDE

Paste Volume	Avg. Voids Area / Joint ($\mu\text{m}^2/\text{joint}$)			
	Corner		Side	
	Component 1	Component 2	Component 1	Component 2
High	15,938	12,580	18,769	13,767
Medium	10,716	9,569	13,796	10,894
Low	1,683	1,919	1,672	1,777

Based on the preceding results, it could be stated that the side contains more voids than the corner. However, it cannot be generalized since certain combinations have some differences. Another conclusion drawn from these results is that the solder paste volume has the same impact as described in section 3.1.

E. Analysis of Variance

ANOVA has been conducted to validate the influence of the stated factor on the voids area. In addition, ANOVA provides the capacity to comprehend and define the interconnections between these factors. The following four variables were examined:

- Solder paste alloys: Three grades: SAC305, M758, and M794.
- Solder paste/sphere ratio: Three levels: low, medium, and high.
- Surface finishes: Two levels: OSP and ENIG.
- Region: Two levels: side and corner.

Fig. 14 (a) depicts the effect of Paste Alloy, Paste Volume, Surface Finish, and Solder Joint Location on the voids area. M794 had the poorest performance, whereas SAC305 performed the best. The paste volume had the same impact as illustrated in section 3.1. The OSP surface finish had more voids area than ENIG, and the surface finish effect was slightly more significant than the solder joint location, which had no influence on the voids area. The interactions between paste alloy and paste volume, as well as paste alloy and surface finish, were very significant. A high volume of M794 combined with ENIG would be the worst of all examined combinations.

IV. CONCLUSION

The impacts of SAC305, M758, and M794 alloys with distinct surface finishes, namely ENIG and OSP, as well as solder joint location, either corner or side, were examined in this study. ANOVA tests were performed to validate the following findings:

- Solder paste volume considerably impacts solder voiding because the larger the volume, the larger the total voids area.
- The total voids area varies significantly with alloy. M794 has much larger voids than other alloys.
- OSP joints have larger voids than ENIG joints for SAC305 and M758, while ENIG is at the top with M794.
- The location of the solder joint does not influence the voids area.
- The worst combination in terms of total voids area was joints printed with a high volume of M794 with ENIG surface finish.

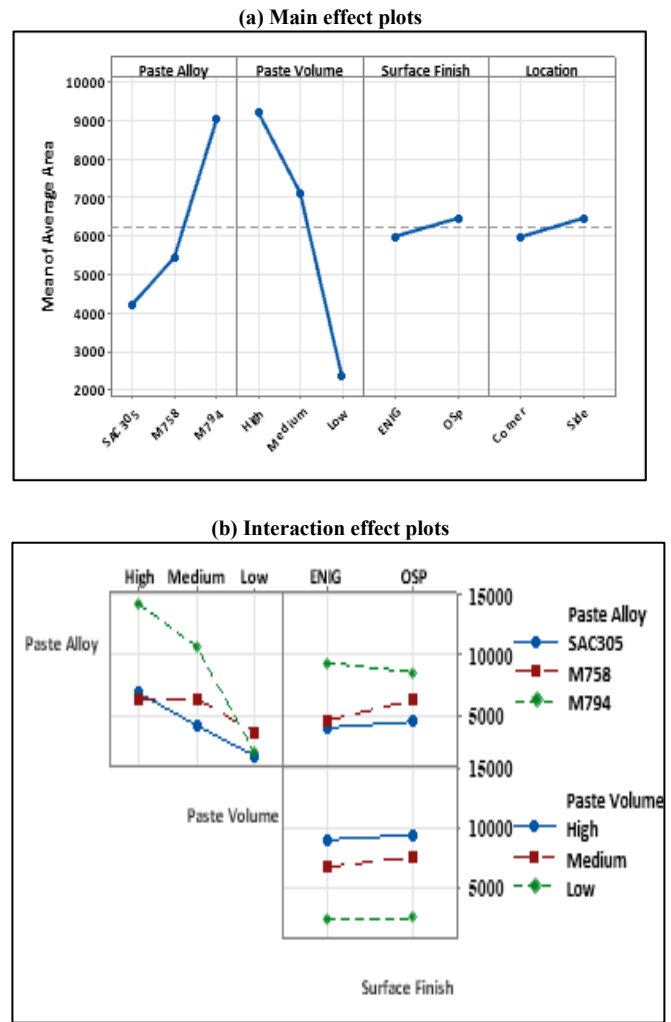


Fig. 14. ANOVA test outcomes; (a) main effect plots, and (b) Interaction effect plots

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